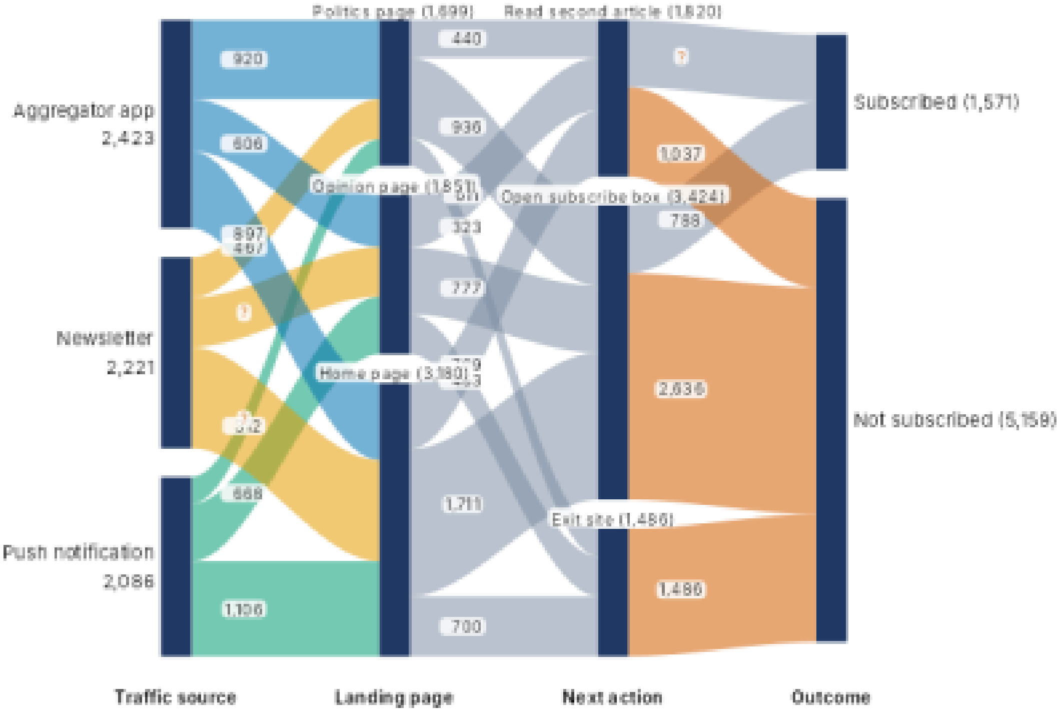
## DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

**R.No: 2 Reg. No: 192424331 Name: AKILA S**

*Scenario: The analytics team of a news portal tracked 6,730 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".*



**(a)**

Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Newsletter to Home page; Read second article to Subscribed; Newsletter to Opinion page. State the rule you used. [4]

**(b)** Compute: (i) the overall conversion rate, i.e. Subscribed as a percentage of all visits; (ii) the largest single flow into "Not subscribed" or "Exit site"; (iii) the percentage of "Home page" visits that go on to "Read second article". [4]

**(c)** Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

**(d)** The Sankey is to be printed with 1 cm of node height = 300 visits. Compute the height in cm of each landing-page node (Home page, Opinion page, Politics page). [2]

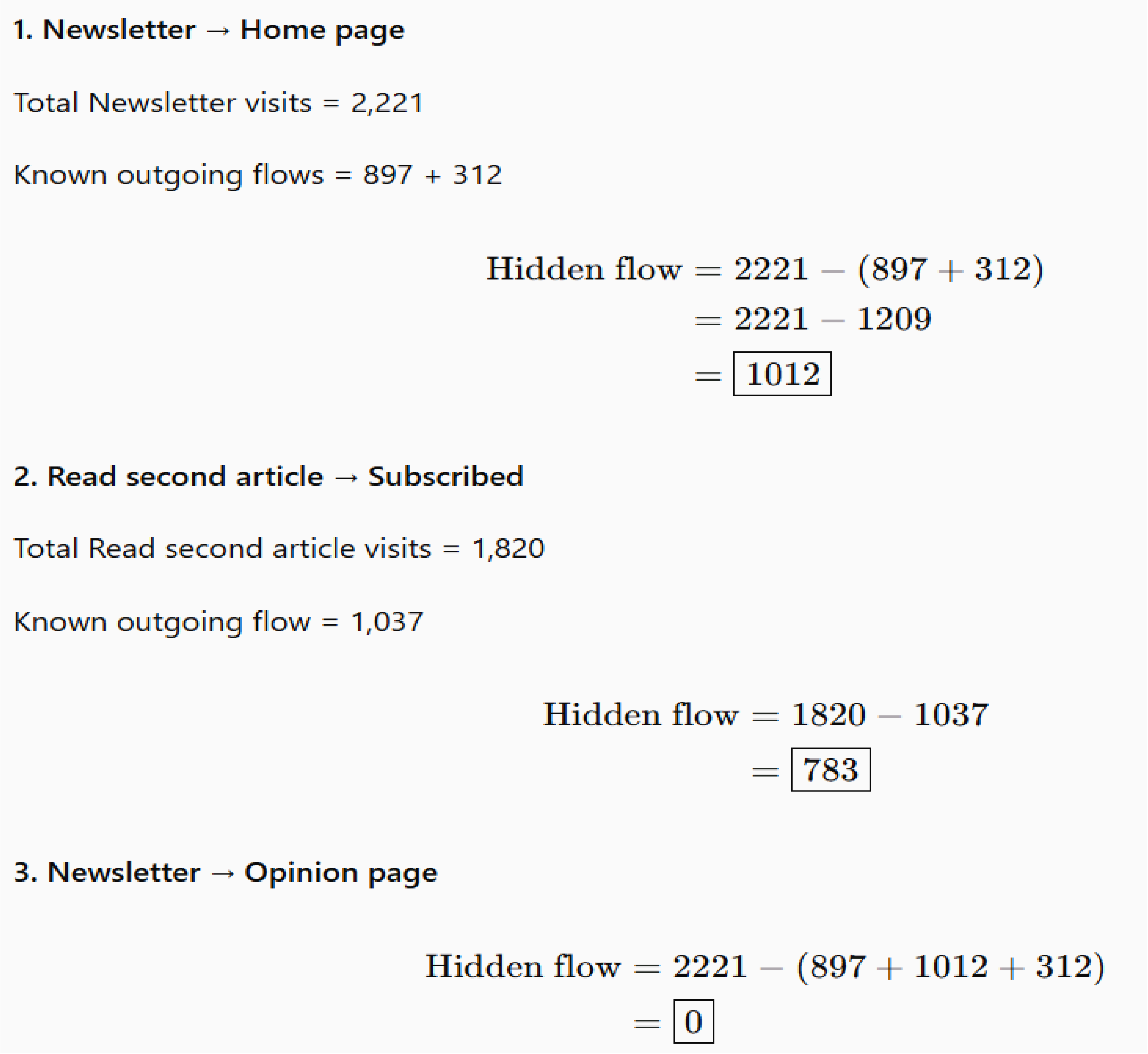
**(e)** Write R code: a links data frame for the first four flows leaving "Newsletter" and "Push notification" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

**(f)** State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

**SOLUTION:**

**(a) Find the three hidden flows using conservation of flow [4 Marks]**

Rule: Total flow entering a node = Total flow leaving the node.



**1. Newsletter → Home page**

Total Newsletter visits = 2,221

Known outgoing flows = 897 + 312

$$\begin{aligned}\text{Hidden flow} &= 2221 - (897 + 312)\\ &= 2221 - 1209\\ &= \boxed{1012}\end{aligned}$$

**2. Read second article → Subscribed**

Total Read second article visits = 1,820

Known outgoing flow = 1,037

$$\begin{aligned}\text{Hidden flow} &= 1820 - 1037\\ &= \boxed{783}\end{aligned}$$

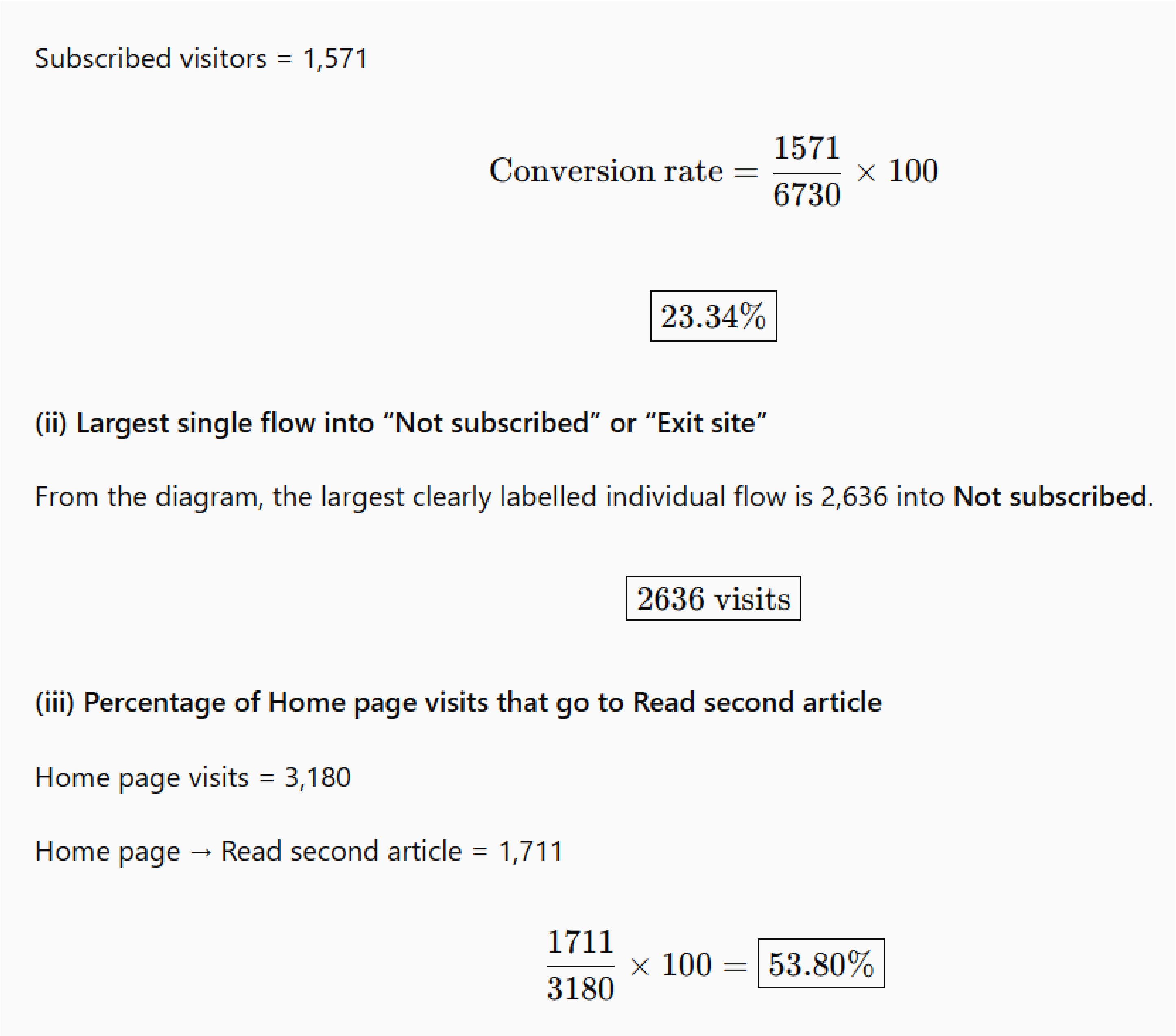
**3. Newsletter → Opinion page**

$$\begin{aligned}\text{Hidden flow} &= 2221 - (897 + 1012 + 312)\\ &= \boxed{0}\end{aligned}$$

## (b) Calculations [4 Marks]

(i) Overall conversion rate

Total visits = 6,730



Subscribed visitors = 1,571

$$\text{Conversion rate} = \frac{1571}{6730} \times 100$$

$$\boxed{23.34\%}$$

### (ii) Largest single flow into “Not subscribed” or “Exit site”

From the diagram, the largest clearly labelled individual flow is 2,636 into **Not subscribed**.

$$\boxed{2636 \text{ visits}}$$

### (iii) Percentage of Home page visits that go to Read second article

Home page visits = 3,180

Home page → Read second article = 1,711

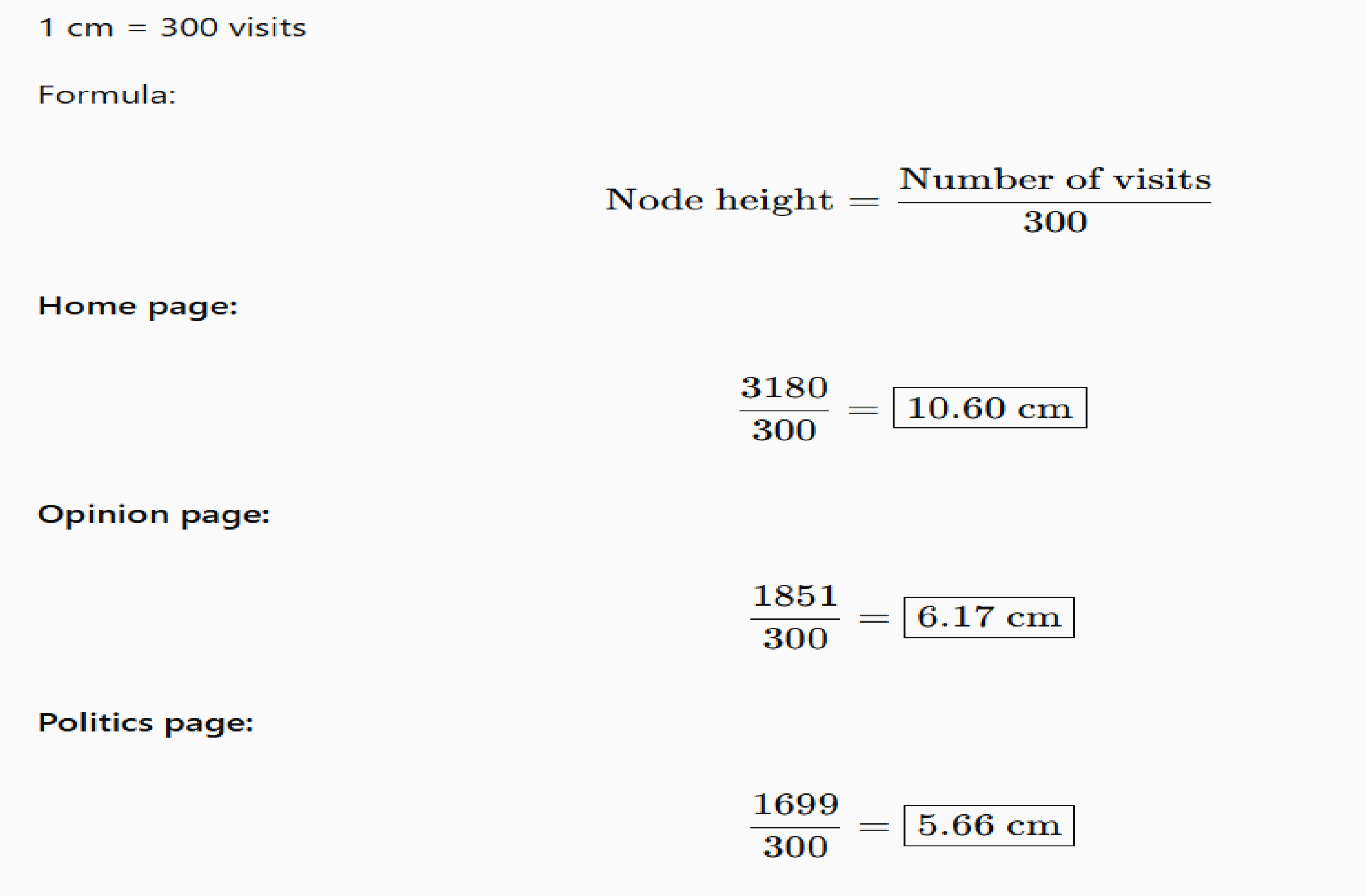
$$\frac{1711}{3180} \times 100 = \boxed{53.80\%}$$

**(c) Four design decisions for the final Sankey diagram [4 Marks]**

1. Node order: Arrange nodes in four columns—Traffic source, Landing page, Next action and Outcome—to maintain a clear reading direction and reduce crossings.
2. Colour rule: Use distinct colours for traffic sources and consistent colours for each source's outgoing flows to make paths easy to trace.
3. Labels: Display node names, visit counts and flow values clearly, avoiding overlapping text and ensuring hidden values are calculated before printing.
4. Highlighting: Emphasize the Subscribed outcome with a contrasting colour to make the conversion path easy to identify.

**(d) Calculate the landing-page node heights [2 Marks]**

Given:



1 cm = 300 visits

Formula:

$$\text{Node height} = \frac{\text{Number of visits}}{300}$$

**Home page:**

$$\frac{3180}{300} = \boxed{10.60 \text{ cm}}$$

**Opinion page:**

$$\frac{1851}{300} = \boxed{6.17 \text{ cm}}$$

**Politics page:**

$$\frac{1699}{300} = \boxed{5.66 \text{ cm}}$$

**(e) R code to create the Sankey diagram [4 Marks]**

```
library(networkD3)

# Links data frame
links <- data.frame(
  source = c(0, 0, 1, 1, 2, 2),
  target = c(3, 4, 3, 4, 3, 5),
  value  = c(897, 312, 1012, 897, 668, 1106)
)

# Nodes data frame
nodes <- data.frame(
  name = c(
    "Newsletter", "Push notification", "Aggregator app",
    "Home page", "Opinion page", "Politics page"
  )
)

# Draw Sankey diagram
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 25)
```

**(f) Two limitations of Sankey diagrams and alternatives [2 Marks]**

1. Too many nodes and crossing flows: The diagram becomes cluttered and difficult to read. Alternative: Use a matrix heatmap.
2. Very small flows become difficult to see: Minor pathways may be hidden by larger flows. Alternative: Use a bar chart to compare individual flow values.